

Pratyush Agarwal

hi@pratyushagarwal.com | +1 (217) 305-3172 | [LinkedIn](#) | [GitHub](#) | [pratyushagarwal.com](#)

SUMMARY

Software Engineer building full-stack and AI-powered products, backed by **two years of production Data and Cloud Infrastructure experience** across production pipelines, APIs, and LLM systems (Python, TypeScript, FastAPI, Claude API, MCP, LangChain, LangGraph).

EXPERIENCE

Archer-Daniels-Midland Co. | Software Engineer Intern | Champaign, IL

Aug 2026 – Present

- Automated multiple workflows with Office Scripts (TypeScript), VBA, SharePoint, and Excel macros to replace manual data cleaning and reporting from lab experiments, with early results indicating **approximately 15 hours/week in researcher time savings**.
- Prototyped a Python model that predicts **extrusion experiment outcomes** from input parameters, helping researchers target promising parameter ranges and cutting failed runs; early results point to a **20% reduction in trial-and-error**.

Kohler Co. | Software Engineer Intern | Champaign, IL

Jun 2025 – May 2026

- Built and shipped production data services integrating **8 source systems**, writing PySpark and orchestration code that **powered company-wide analytics and AI automation initiatives**.
- Optimized pipeline performance through partitioning and incremental loads, **cutting data ingestion time by 40%** and **query execution time by 25%** on high-volume tables.

Apna | Data Engineer | Bangalore, India

Oct 2022 – Aug 2024

- Built a user verification and data ingestion service using SQL, Python, Postman, and Government of India APIs, **cutting processing time for Operations agents by around 50%** and improving onboarding for **roughly 1,200 users** weekly.
- Saved around \$2,500/month in BigQuery warehouse spend** and **cut query execution time by roughly 20%** through partition-filtering and query scope-limiting on high-cost workloads.
- Partnered with Trust and Safety leadership and operations managers to consolidate fragmented reactive and proactive investigation workstreams into a single unified system, **reducing investigation overlap by around 35%** and **saving 20+ agent hours weekly**.

Google | Strategy and Analytics Intern | Singapore

Oct 2021 – Apr 2022

- Engineered web scraping scripts with ChromeDriver and Selenium to capture daily metrics across **100+ retailer websites**, mining page archives for backfill data and powering **13 dashboards** across brand prominence, pricing, inventory, and delivery timelines.
- Pitched and shipped a **brand prominence algorithm** scoring each brand's **share of homepage space** across retailer sites, adding a new tracking pillar that fed leadership dashboards during the Pixel 6 launch.

PROJECTS

Mmemo: Personal Knowledge Engine | TypeScript, NestJS, Python, FastAPI, PostgreSQL, Docker

[\[GitHub\]](#)

- Building a two-service "second brain" app: a TypeScript/NestJS API spanning **4 core modules** (notes, tasks, auth, kanban) and a Python/FastAPI AI service communicating over an HTTP contract, orchestrated with Docker Compose.
- Architected with clean layering (thin routers, service logic, repository DB access) on Postgres and pgvector, targeting **sub-second semantic search** and evolving toward GraphRAG over Neo4j and an MCP-exposed agent.

TickerSense | React, Next.js, Node.js, Claude API, PostgreSQL

[\[GitHub\]](#)

- Built a full-stack LLM platform with React, Next.js, and Node.js unifying **4 SEC filing types** (10-K, 10-Q, 8-K, DEF 14A) and **5 years** of financials into one interface; placed **2nd at the Claude Builder Club Hackathon (UIUC)**.
- Developed a source-grounded research assistant using the Claude API with year-over-year risk analysis across filings and exportable PDF reports.

TripTok | FastAPI, PySpark, AWS, Python

[\[GitHub\]](#)

- Architected an end-to-end pipeline processing **25GB+ (1B+ rows)** of NYC Taxi data with PySpark on AWS S3, partitioned and queried via Amazon Athena, serving ML models through a FastAPI MCP server with AWS Bedrock and Strands Agents.
- Developed **Fare Prediction** and **Travel Time** models (XGBoost, Random Forest, LSTM) behind a conversational LLM analytics layer for real-time mobility insights.

TECHNICAL SKILLS

Programming Languages: Java, Python, TypeScript, JavaScript, SQL, Bash/Shell Scripting

AI and Machine Learning: LLM Applications (Claude API, Cursor, Strands), LangChain, LangGraph, Agents, MCP, RAG, Prompt Engineering, Model Evaluations, Pandas, NumPy, scikit-learn

Frontend and Backend: Node.js, NestJS, FastAPI, REST APIs, GraphQL, SQLAlchemy, Postman, Selenium, pytest, Jest, React.js, Next.js, HTML, CSS

Cloud and DevOps: AWS (EC2, S3, Athena, Bedrock, Lambda, API Gateway, IAM), Azure, GCP, Docker, Kubernetes, Git/GitHub, CI/CD, Microservices, Agile/Scrum

Databases and Data: PostgreSQL, DynamoDB, Redis, BigQuery, MySQL, SQL Server, Neo4j, pgvector, Databricks, PySpark, Parquet, ETL, Power BI

CS Foundations: Data Structures, Algorithms, Object-Oriented Programming, System Design

EDUCATION

University of Illinois at Urbana-Champaign

Aug 2024 – May 2026

Master of Science in Information Management; GPA: 3.75/4.0

Champaign, IL

Relevant Coursework: Applied Machine Learning, Cloud Computing Applications (8 AWS Mini Projects), Data Mining, Data Warehousing and Business Intelligence.

Certifications: LangChain Certified Agent Engineer.

Manipal University Jaipur

Jul 2018 – Jul 2022

Bachelor of Technology in Computer and Communication Engineering; GPA: 8.53/10.0

Jaipur, India

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming in Java, Relational Database Management Systems, Operating Systems, Computer Networks, Software Testing, Cryptography and Security.

Certifications: Certified Ethical Hacker (EC Council), Networking in Google Cloud, Architecting with Google Kubernetes Engine, Google IT Support.